

Enhancing Disease Detection in South Asian Freshwater Fish Aquaculture Through Convolutional Neural Networks

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1. Introduction

Aquaculture is one of the fastest-growing food production sectors in South Asia, and disease outbreaks are among its costliest threats. Identifying sick fish early still depends heavily on visual inspection by aquatic animal health experts, which is slow, subjective, and does not scale. This project investigates whether convolutional neural networks (CNNs) can automate that task: given a photograph of a fish, classify it as healthy or as suffering from one of six common diseases.

2. Dataset

The dataset comprises 1,747 images of South Asian freshwater fish across seven classes:

- Bacterial red disease
- Bacterial diseases - Aeromoniasis
- Bacterial gill disease
- Fungal diseases - Saprolegniasis
- Healthy fish
- Parasitic diseases
- Viral diseases - White tail disease

Images were collected from public sources (Kaggle). The data were split into training, validation, and test sets, with 697 images held out for final testing.

3. Methods

Three modeling approaches were compared, in order of development:

- Baseline CNN (CNN.ipynb): a convolutional network trained from scratch - an early experiment that underfit, reaching only ~40% validation accuracy.
- Transfer learning (TransferLearning_ImageNet.ipynb): an ImageNet-pretrained backbone fine-tuned on the fish dataset, reaching ~71% test accuracy.
- FishNet CNN (Fishnet_cnn.ipynb): the final architecture - convolutional layers for feature extraction, max-pooling for downsampling, dense layers for classification, and dropout for regularization. Trained with categorical cross-entropy loss and the Adam optimizer, monitoring training and validation performance throughout.

4. Results

The final FishNet model achieved 92.5% accuracy on the 697-image test set (test loss 0.29), with macro-averaged precision, recall, and F1-score of 0.93. Per-class performance:

Class	Precision	Recall	F1-score	Support
Bacterial red disease	0.88	0.94	0.91	100
Aeromoniasis	0.96	0.90	0.93	100
Bacterial gill disease	0.95	0.93	0.94	100
Saprolegniasis (fungal)	0.96	0.93	0.94	100
Healthy fish	0.93	0.97	0.95	100
Parasitic diseases	0.84	0.97	0.90	100
White tail disease (viral)	0.99	0.84	0.91	97

Training-history analysis showed accuracy rising quickly, then plateauing with signs of eventual overfitting - the expected pattern for a moderately sized image dataset trained without heavy augmentation.

5. Discussion

The progression across the three approaches tells the story: a from-scratch CNN could not learn robust features from 1,747 images alone, transfer learning helped substantially, and the tuned FishNet architecture delivered production-grade accuracy. The model performs strongly across all disease categories, with the viral white-tail class showing the highest precision (0.99) and parasitic diseases the highest recall (0.97). Future improvements include stronger regularization and data augmentation for better generalization.

6. Conclusion

CNNs offer a practical path to automated disease detection in freshwater aquaculture, reducing reliance on scarce expert inspection and enabling timely intervention to limit infection spread and economic losses. The final model classifies seven fish health conditions at 92.5% test accuracy, supporting real-time surveillance and adaptive management strategies for more sustainable aquaculture.

7. Presentation

This work was presented as a research poster at Arkansas Tech University: "Enhancing Disease Detection in South Asian Freshwater Fish Aquaculture Through Convolutional Neural Networks" (48x36 poster, included in this repository).